



CS & IT ENGINEERING



Algorithms

Divide & Conquer

Lecture No.- 02



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Intro to Divide & Conquer

Min-Max Algo (Non D_nC)

Topics to be Covered



Topic

Topic

Topic

DaC of Min-Max Algo
Binary Search



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe



Topic : Min-Max Problem

Divide & Conquer based

Total number of comparisons (Algo2)

Best $\rightarrow \underline{(n - 1)}$

Worst $\rightarrow \underline{2*(n-1)}$

Avg $\rightarrow \underline{\left(\frac{3n}{2} - 1\right)}$ ☆

IF ✓
IFX

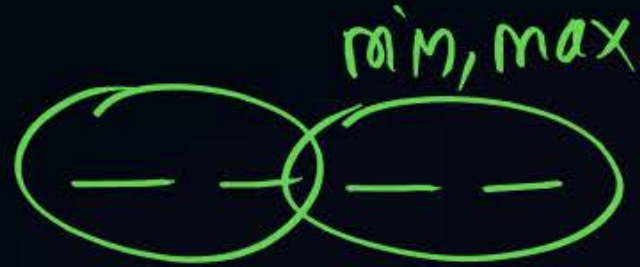
PnC based Min-Max



Topic : Min-Max Problem

Divide & Conquer based

(min-max)



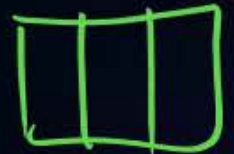
1	2	3	4	5	6	7	8	9
7	11	-1	-6	14	19	2	63	8

$$\frac{1+9}{2} = 5$$

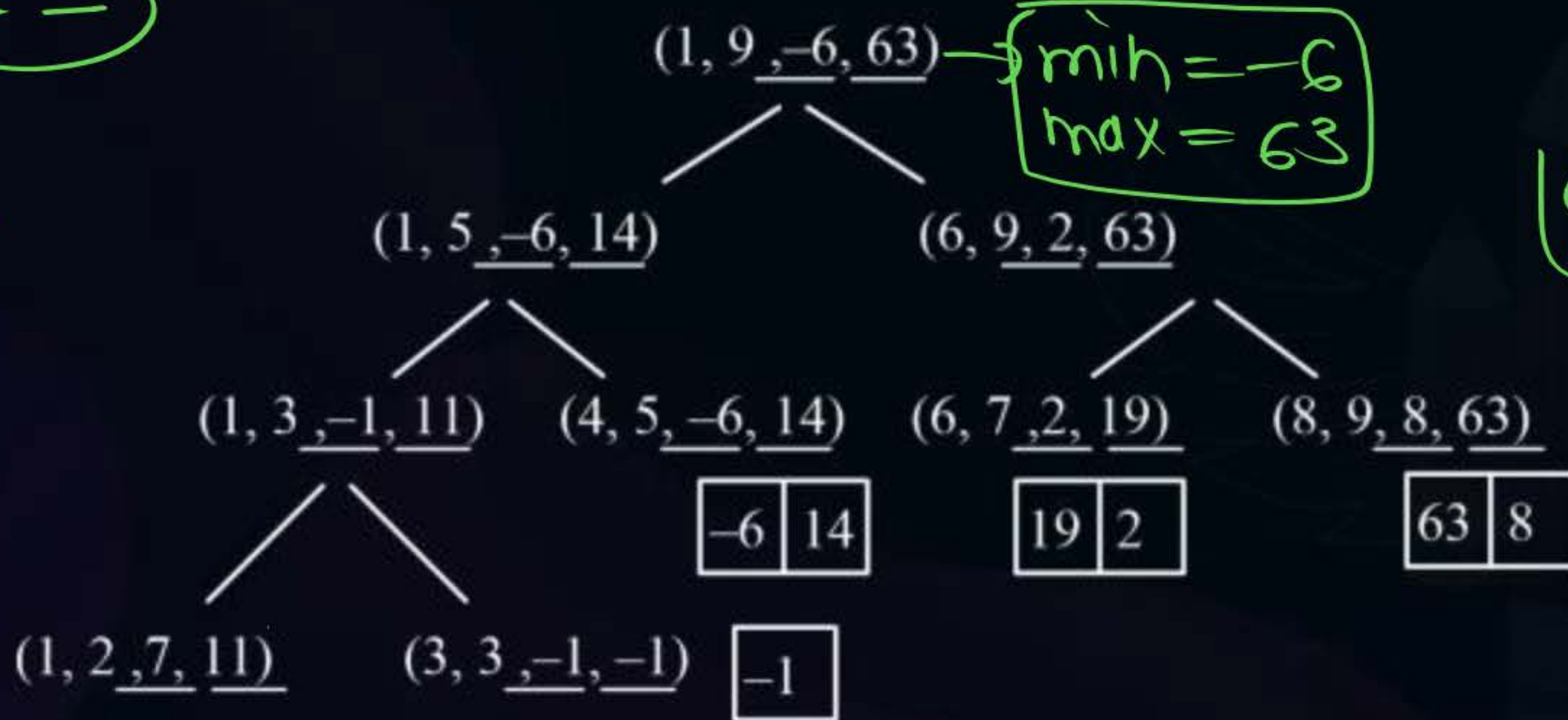
min = -6
max = 63

$$\left\lfloor \frac{6+9}{2} \right\rfloor = 7$$

$$\frac{1+5}{2}$$



$$\frac{1+3}{2}$$

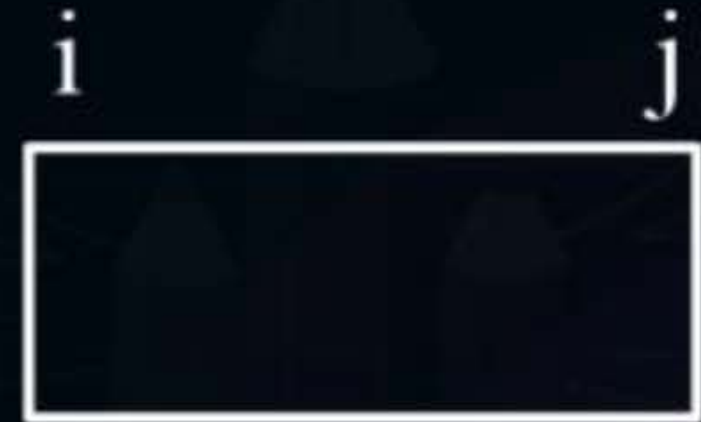




Topic : Min-Max Problem



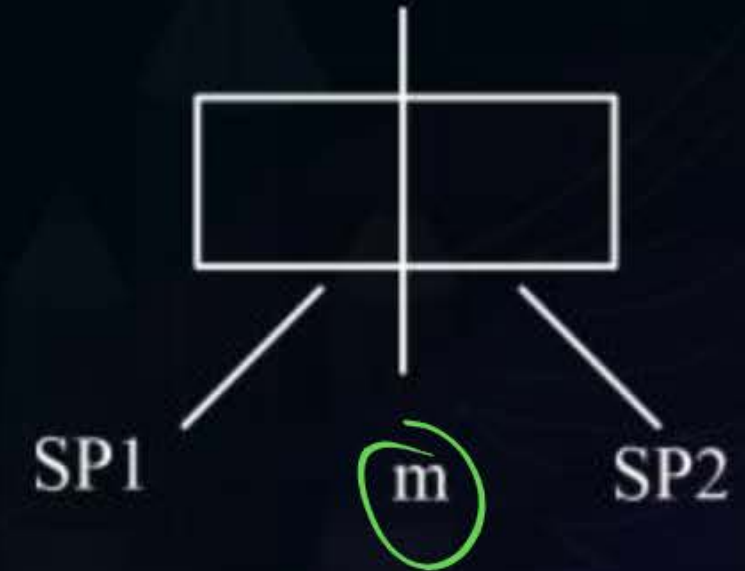
1. Algorithm $\text{MaxMin}(i, j, \text{max}, \text{min})$
2. //a[1: n] is a global array. Parameters i and j are integers.
3. {
4. If $(i == j)$ the max: min: = $a[i]$; // Small (P)
5. else if $(i \leq j - 1)$ then // Another case of Small (P)
6. {
7. If $(a[i] < a[j])$ then
8. {
9. $\text{Max} := a[j]$; $\text{min} := a[i]$;
10. }





Topic : Min-Max Problem

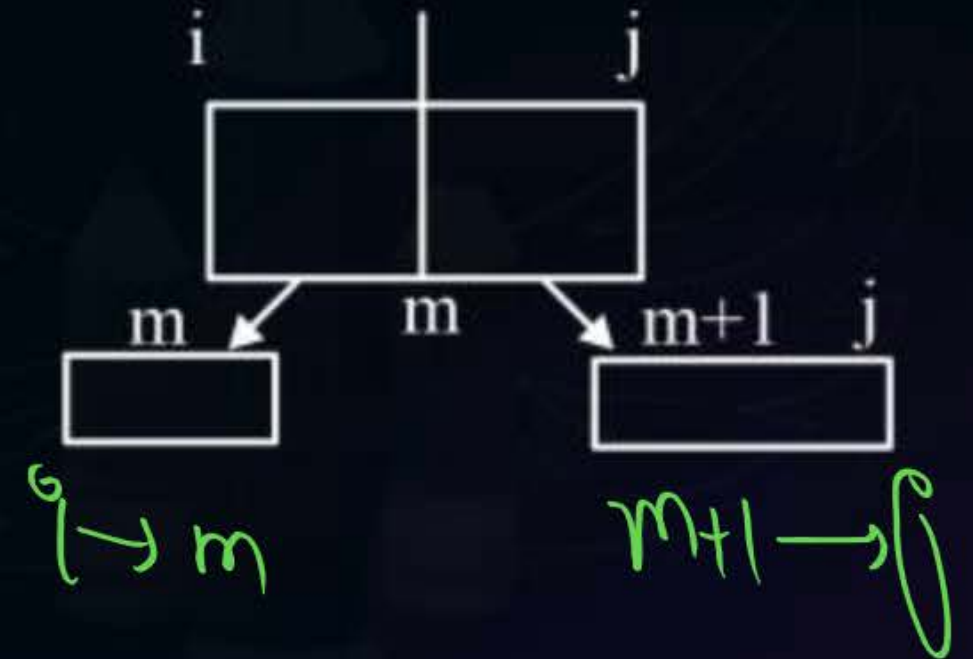
```
13.     else
14.     {
15.         max := a[i]; min := a[j]
16.     }
17. }
18. → else → large Problem
19. { // If P is not small, divide P into subproblems.
20.     // Find where to split the set.
```





Topic : Min-Max Problem

21. $\text{mid} = \left\lfloor \frac{i+j}{2} \right\rfloor$
22. // Solve the subproblems.
23. MaxMin (i, mid, max, min);
24. MaxMin (mid + 1, j, max1, min1);
25. // Combine the solution.
26. If ($\text{max} < \text{max1}$) then $\text{max} := \text{max1}$; — ①
27. If ($\text{min} > \text{min1}$) then $\text{min} := \text{min1}$; — ①
28. }
29. }





Topic : Min-Max Problem

Performance of Divide & Conquer based

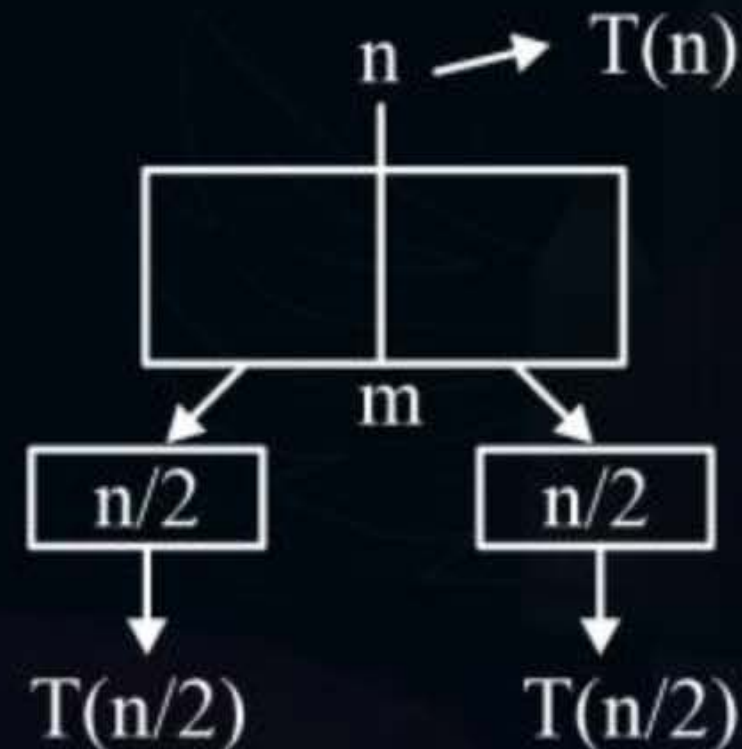
Number of element comparisons:

Let $T(n) \rightarrow$ number of elem comparison reqd. for min-max on 'n' elem.

$$T(n) = 0, n = 1$$

$$T(n) = 1, n = 2$$

$$T(n) = 2T(n/2) + 2, n > 2$$





Topic : Min-Max Problem

#Q. $T(n) = 2T(n/2) + 2$
 $T(n/2) = 2T(n/2^2) + 2$
 $T(n) = 2[2T(n/2^2) + 2] + 2$
 $= 2^2T(n/2^2) + 2^2 + 2$
 $= 2^2[2T(n/2^3) + 2] + 2^2 + 2$
 $= 2^3T(n/2^3) + (2^3 + 2^2 + 2^1)$

A

$(n/2 + 1)$

B

$(n-1)$

C

$\left(\frac{3n}{2} - 1\right)$

D

$\left(\frac{3n}{2} - 2\right)$

General then

$$T(n) = 2^k T(n/2^k) + (2^k + 2^{k-1} \dots 2^1)$$

$$\Rightarrow T(n) = 2^k T(n/2^k) + \sum_{i=1}^k 2^i$$

value ?

✓
 $[2^{k+1} - 1]$



Topic : Min-Max Problem

$$2^1 + 2^2 + \dots + 2^k$$

$$GP: \left. \begin{array}{l} a = 2 \\ r = 2 \\ n = k \end{array} \right\} \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{2(2^k - 1)}{2 - 1} = \boxed{2 \times 2^k - 2}$$

$$[2 \times 2^k - 2]$$



Topic : Min-Max Problem

$$T(n) = 2^k T(n/2^k) + 2(2^k - 1)$$

For BC

$$\cancel{n/2^k = 2} \quad n/2^k = 2$$

$$n = 2 \times 2^k$$

$$n = 2^{(k+1)}$$

$$(k+1) = \log_2 n$$

$$\Rightarrow \frac{n}{2} \times T(2) + 2\left(\frac{n}{2} - 1\right)$$

$$\Rightarrow \frac{n}{2} \times 1 + n - 2$$

$$\Rightarrow \frac{n}{2} + n - 2 = \boxed{\frac{3n}{2} - 2} \quad \checkmark$$



Topic : Min-Max Problem

Summary: DnC vs D&C → Min-Max Algo:

V.V. Imp

	<u>Non-DnC</u> (Algo2)	<u>DnC</u>	
Best case Algo2 Deer order	$(n-1)$	$\left(\frac{3n}{2} - 2\right)$	
Wrost case Algo2 Iner order	$2*(n-1)$	$\left(\frac{3n}{2} - 2\right)$	★
Avg case (Random order)	$\left(\frac{3n}{2} - 1\right)$	$\left(\frac{3n}{2} - 2\right)$	

↓
always



Topic : Min-Max Problem

Space complexity:-

Non-D&C (Algo2) $\rightarrow 0(1)$

D&C $\rightarrow$ Recursion stack $\rightarrow 0(\log_2 n)$

1

TC: non DnC $\rightarrow 0(n)$

DnC $\rightarrow 0(n)$





Topic : Binary Search



Binary Search:

1

Binary Search



Topic : Binary Search



#Q. Given a Binary Tree of 'n' nodes,
min. Height (depth)
max. Height (depth)

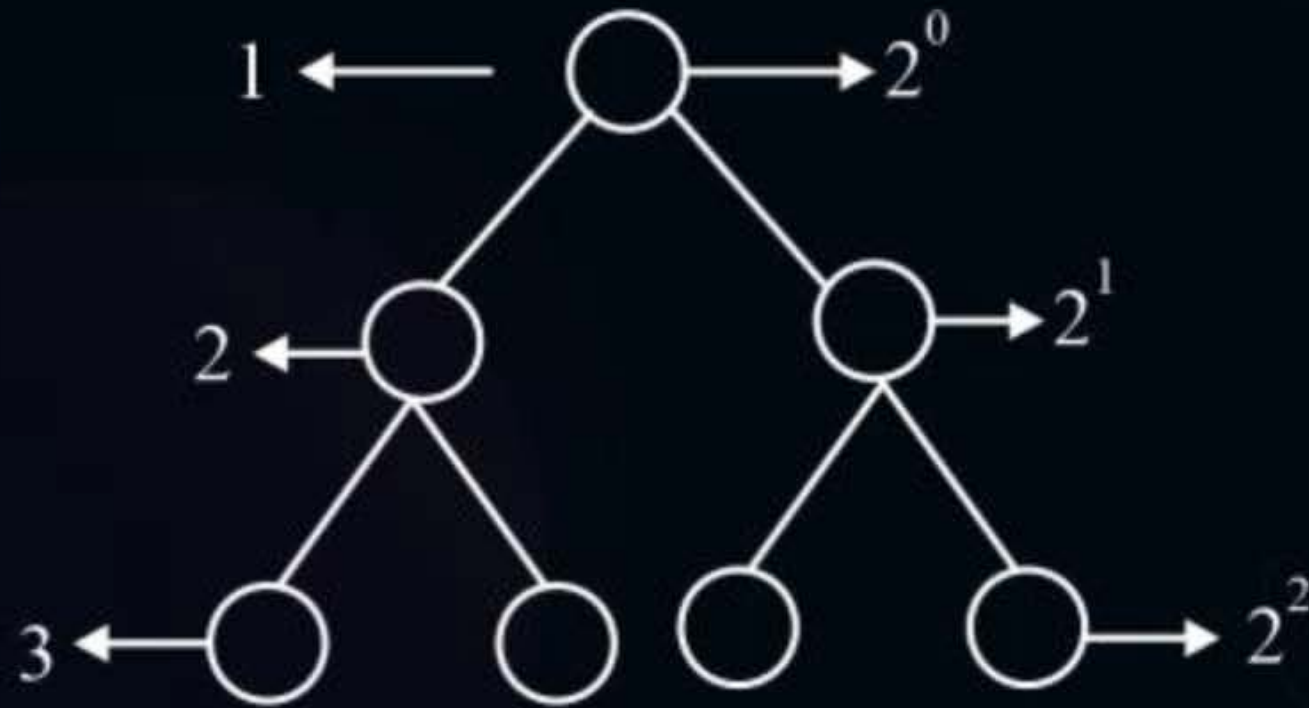




Topic : Binary Search



Level:-



Complete
BT



Topic : Binary Search



Max number of nodes at any level 'i' in BT = $2^{(i-1)}$



$$n = 50$$

$$1 \text{ box} = \text{max } 10$$

$$\boxed{10} \quad \boxed{10} \quad \dots \quad \boxed{10}$$

5 boxes



Topic : Binary Search

#Q. Total number nodes ^{n} ~~in~~ a full Binary Tree of height h .

$$= 2^{1-1} + 2^{2-1} + 2^{3-1} \dots + 2^{h-1}$$

$$= \sum_{i=1}^h 2^{(i-1)} = \sum_{i=1}^h \frac{2^i}{2} = \frac{1}{2} * \sum_{i=1}^h 2^i$$

Total nodes:

$$n = \frac{2(2^h - 1)}{2 - 1} \times \frac{1}{2}$$

$$n = 2(2^h - 1) \times \frac{1}{2}$$

$$n = 2^h - 1$$

$$2^h = n + 1$$

$$\underline{h = \log_2(n+1)} \approx \underline{h = O(\log_2 n)}$$

$\rightarrow$ min Height



Topic : Binary Search



Level:

every level 1 elem

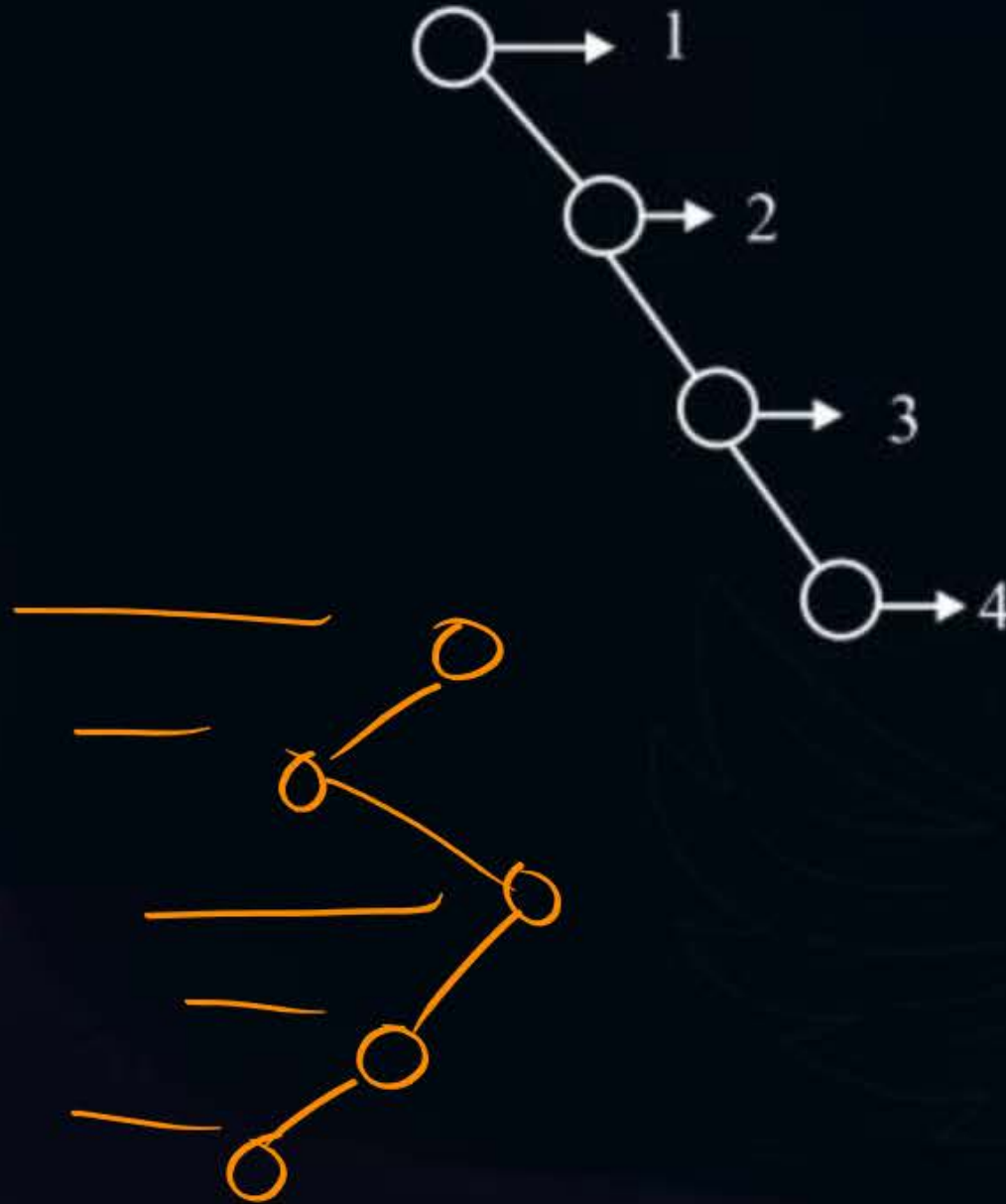
Height h

Total nodes = $\sum_{i=1}^h 1$

$$n = \frac{(1 + 1 + \dots 1)}{h \text{ times}}$$

$$n = h \Rightarrow h = n$$

Max-height of a BT





Topic : Binary Search



Range of height of a Binary Tree.

Level start at 1

$$\log_2^n \leq h \leq n$$





Topic : Min-Max Problem

#Q. Given a Binary Tree

level starts at 1

if every ~~every~~ level i has exactly 'i' nodes

Height of such a BT = ?

33.3%

A

n^2

B

$\sqrt{n}$

C

n

D

$\log n$



Total no. of nodes = $1 + 2 + 3 + 4 + \dots + h$

$$n = 1 + 2 + 3 + \dots + h$$

$$n = \frac{h(h+1)}{2}$$

$$\Rightarrow \frac{h^2 + h}{2} = n$$

$$\Rightarrow h^2 + h = 2n$$

$$h^2 \approx 2n \Rightarrow h = O(\sqrt{n})$$

V. Imp:

$$A = \begin{bmatrix} 20 & 72 & 900 & 300 & 742 & 9999 & 893 & 542 & 5 & 1200 \\ 2 & 3 & & 4 & 6 & & & 5 & 1 & \end{bmatrix}$$

(8) If Radix Sort is used to Sort above array in asc order, then after 4th pass, which will be the 6th elem?

$$\text{No. of passes} = \log_2(9999) = 4$$

Ans = 742



2 mins Summary



Topic

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Topic

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Min-Max

Binary Search



THANK - YOU